

Identification of Novel Cryptic Pockets in RAC1

Elvis A. F. Martis and Stéphane Téletchéa*

US2B, Nantes Université, CNRS, UMR 6286, F-44000 Nantes, France

E-mail: stephane.teletchea@univ-nantes.fr

Abstract

The work described here attempts to locate cryptic pockets in the RAC1 protein.

More to be written

Introduction

- importance of GTPases, their switch mechanism and their interactions with GEFs, GAPs and GDIs Small GTPases act as molecular switches cycling between OFF and ON states. These switching mechanism thus controls a vast variety of processes that control cell growth and differentiation to vesicular and nuclear transport.
- structural biology information about RAC1 and protein-protein interactions with GEFs, GAPs and GDIs
- Therapeutic interventions on RAC1 and their disadvantages
- Binding site information and changes in the binding sites, if any are reported
- Are there any reports on cryptic pockets for RAC1

Methods

- List of PDBS used
- Protocol for AMBER MD simulations Preparation of systems for molecular dynamics simulation The coordinates for the protein-ligand complexes were retrieved from the Protein Data Bank (PDB); the PDB entries used in this study are listed in Table S1 in the supplementary information. The X-ray structures were imported and processed in the Maestro-GUI (Schrödinger suite, 2018). All systems were prepared for MD simulations with the Leap module in AMBERtools15. Protein atoms were atom-typed according to the AMBER14 (ff14SB) force field. Generating AMBER-compatible parameters for ligands.

Molecular dynamics simulations All complexes were solvated with TIP3P waters²² enclosed in a box with dimensions extending 12 Å beyond any protein atom, and an appropriate number of neutralizing ions were added to maintain the neutrality of the system. The complexes were relaxed using three cycles of minimization of 50,000 steps each. The first 10,000 steps of minimization adopted the steepest descent algorithm, and the remaining 40,000 steps used the conjugate gradient algorithm. The cut-off for evaluating non-bonded interactions was placed at 8.0 Å for the minimization phase. The first cycle of minimization was performed with 25 kcal / (mol Å²) restraining force on the solute atoms, this force constant was reduced to 5 kcal / (mol Å²) in the next cycle. After heating, the system was equilibrated by MD simulations under the NVT ensemble for six cycles so that the cut-off for evaluating non-bonded interactions was placed at 9 Å. In the production phase of the MD simulation, the range of electrostatic interactions were recalculated using the Particle Mesh Ewald (PME) summation method enabled code of PMEMD in AMBER22. All simulations were performed with the SHAKE25 algorithm. The root-mean-square deviation (RMSD) for the protein was computed using the X-ray conformation as the reference. The convergence of MD trajectories was estimated using the root-mean-square average correlation (RAC) with mass-weighted RMS averaging of Cα atoms, with the

mean-square fluctuations (RMSF) per residue for the backbone and side-chain atoms were computed

- - pocket identification
 - pocket volume measurements

fragment placement and trying to keep the pocket open to target drug-like molecules

- Type of pocket and the type of fragment more suitable to keep it open

Analysis of other MD observations to support pocket volume changes

- RMSDs for switch region 1 and 2
- (perhaps) PCA analysis to explain the switch region dynamics and the pocket volumes

Results and discussion

- Try to look for a connection/relation between the following
 - switch region dynamics with different binding partners
 - switch region dynamics with pocket volumes
 - If we find different pockets that might open and close with different binders

TOC Graphic

Some journals require a graphical entry for the Table of Contents. This should be laid out “print ready” so that the sizing of the text is correct. Inside the tocentry environment, the font used is Helvetica 8 pt, as required by *Journal of the American Chemical Society*.

The surrounding frame is 9 cm by 3.5 cm, which is the maximum permitted for *Journal of the American Chemical Society* graphical table of content entries. The box will not resize if the content is too big: instead it will overflow the edge of the box.

This box and the associated title will always be printed on a separate page at the end of the document.